



LEVEL EDITING

A complete practical tutorial for building a playable environment with every Level Design panel in 3DG Engine.

PROJECT

Moonwatch Courtyard

OUTCOME

A measured, layered, validated courtyard with modular architecture, a spline road, fences, props, foliage, bookmarks, and saved variants.

EDITOR WORKFLOW GUIDE | AUGUST 2026



TUTORIAL OVERVIEW

01. Plan the level before placing assets

This tutorial uses one compact environment to teach the complete level-editing workflow. Build in passes: shape, measure, assemble, decorate, organize, compare, validate, and save. Do not begin with final props; the tools are most effective when each pass has a clear purpose.

Finished level brief

Zone	Target size	Required result
Arrival road	4 m wide x 18 m long	Spline road with low fences and lantern props
Courtyard	16 m x 14 m	Enclosed room shell, gate opening, centered shrine
Shrine platform	6 m x 5 m	Stairs, aligned columns, radial detail props
Garden edges	2 m border	Scatter-painted rocks, shrubs, and grass
Review states	3 variants	Blockout, dressed, and gameplay-readable versions

Rule of the tutorial

Use meaningful group names from the beginning: MC_Blockout, MC_Road, MC_Fence, MC_ShrineArray, and MC_Room. Generated tools can replace an existing group with the same name, which makes iteration predictable.

Tool sequence

Pass	Tool(s)	Decision
1 - Shape	Blockout, Room Builder	Is the route and scale correct?
2 - Verify	Measurement, Alignment	Do dimensions and spacing read correctly?
3 - Assemble	Modular Placement, Prefab Palette, Array	Can repeated structure be authored quickly?
4 - Guide	Spline Builder	Do paths, fences, and props follow the intended flow?
5 - Dress	Scatter & Paint	Is detail controlled rather than noisy?
6 - Manage	Layers, Bookmarks, Variants	Can the level be reviewed and revised safely?
7 - Ship	Validation & Cleanup	Are broken references and unsafe values resolved?

BEFORE THE FIRST OBJECT

02. Prepare the editor and project

1. Open or create the project, then create a scene inside the project's Scenes folder. Save it as MoonwatchCourtyard.scene.
2. Open Panels > Core and enable Viewport, Hierarchy, Inspector, Assets, and Gizmo. Open the level tools from Panels > Level Design.
3. Create project folders for Architecture, Props, Foliage, Materials, Prefabs, and LevelVariants. Keep engine-owned assets in Content.
4. Import the meshes and materials needed by this exercise. Use engine-owned static mesh, material, and prefab assets so dropdowns in the level tools can discover them.

Recommended starter assets

Type	Examples	Used by
Architecture	wall, column, arch, stair, floor tile	Modular Placement, Array
Props	lantern, crate, bench, statue	Prefab Palette, Spline Builder
Foliage	grass clump, shrub, rock	Scatter & Paint
Materials	stone, plaster, wood, road	Blockout, Room Builder, Spline Builder
Prefabs	lantern post, shrine cluster	Prefab Palette, Modular Placement

Essential viewport controls

Input	Action
W / E / R	Move / rotate / scale gizmo
G	Uniform scale on all axes
Shift + click	Add or remove an object from the current multi-selection
Frame Selected	Bring the selected object into view
Ctrl + S	Save the current scene
Undo / Redo	Revert or restore an editor operation

Capture safety

While viewport camera movement has captured the mouse, panel fields are intentionally inactive. Release camera movement before editing values.

03. Build the first playable blockout

Blockout & Shape Editing Panels > Level Design	PURPOSE Create editable walls, floors, platforms, ramps, doorways, and stairs.	RESULT A fast greybox made from normal scene objects.
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1. Open Blockout & Shape Editing. Set Group Name to MC_Blockout and keep Replace group with same name enabled.
2. Create a floor at the viewport cursor with dimensions 20 x 0.25 x 18 m. Enable a collider and apply a simple stone or neutral blockout material.
3. Create the shrine platform as a Platform with dimensions 6 x 0.6 x 5 m. Place it at the far side of the courtyard.
4. Create a Stairs blockout, set 6 to 8 steps, and size it to the platform height. Rotate with E if the approach faces the wrong direction.
5. Create a Doorway at the courtyard entrance. Start with a 1.8 m opening width and 2.4 m opening height.
6. Use Generate / Update after changing dimensions. Because replacement is enabled, the tool updates MC_Blockout instead of leaving old pieces behind.

Placement modes

Mode	Best use	Watch for
Viewport Cursor	Fast spatial iteration	Confirm the cursor is on the intended surface
Manual	Exact architectural coordinates	Keep units and axis signs consistent
Selected Object	Build relative to an anchor	Select the correct object before generating

Blockout checkpoint

Walk the route in Play mode before dressing. The player must pass through the doorway, climb the stairs, and cross every platform without catching on colliders.

04. Measure and align the composition

Measurement & Ruler Panels > Level Design	PURPOSE Capture persistent distance and box measurements in the viewport.	RESULT Verified widths, heights, clearances, and footprints.
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1. Choose Distance, start capture, then click the two inner doorway edges. Confirm the opening is at least 1.8 m.
2. Choose Box and capture the courtyard footprint. Name it Courtyard Clear Area. Toggle centimeters only when inspecting small prop clearances.
3. Enable grid snapping for exact architectural checks. Hide measurements when presenting the level; they remain editor-only guides.

Object Alignment & Distribution Panels > Level Design	PURPOSE Operate on multi-selected scene objects.	RESULT Clean rows, grids, spacing, and shared edges.
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4. Place four temporary cubes around the shrine. Shift-click them in the desired order; the primary selection acts as the reference when Preserve Primary is enabled.
5. Use axis alignment to give the cubes a shared Y value. Then use Distribute Line with 2 m spacing.
6. Try Arrange Grid with 2 columns and equal X/Z spacing. Keep the result only if it improves the shrine silhouette.

Selection discipline

Alignment edits transforms directly. Keep structural objects on a dedicated layer and save before large multi-selection operations so an accidental selection is easy to undo.

Quality checks

Check	Target
Door clearance	Player capsule plus comfortable side margin
Stair approach	Straight landing with no collider lip
Shrine center	Aligned to the courtyard's visual axis
Traversal lane	No decorative object blocks the shortest route

PASS 1 REFINEMENT

05. Replace the shell with a generated room

Room Builder	PURPOSE	RESULT
Panels > Level Design	Capture two corners and generate a room shell with optional openings.	Editable walls, floor, ceiling, posts, colliders, and material.

1. Hide the temporary outer walls using the Levels layer workflow described later. Keep the floor visible as a reference.
2. Set Room Name to MC_Room, enable Capture Outline, and click opposite corners of the courtyard floor.
3. Use wall height 3.5 m, wall thickness 0.25 m, and floor thickness 0.20 m. Enable colliders and floor generation.
4. Enable the door. Choose the entrance wall, use width 1.8 m, height 2.4 m, and adjust Door Offset until it aligns with the road.
5. Select a wall material and generate. If the doorway is on the wrong wall, change the wall choice and regenerate with replacement enabled.
6. Enable ceiling generation only for an enclosed interior. For this courtyard, leave it off so the sky and outdoor lighting remain visible.

Generated object behavior

Room pieces remain normal scene objects. You can select, rename, move, material-swap, or delete an individual wall after generation. Regenerating with the same room name replaces the group, so make manual exceptions only after the dimensions are final.

When to keep the blackout

Keep the original greybox on a hidden Blockout layer until the generated shell is approved. It gives you a reliable comparison and fallback without cluttering the active viewport.

06. Assemble architecture with modular assets

Modular Placement	PURPOSE	RESULT
Panels > Level Design	Browse engine-owned models and prefabs, then place them with consistent snapping.	A fast kit-based architecture pass.

1. Filter for the wall, column, arch, and floor pieces. Choose a static mesh or prefab, then enable placement.
2. Set grid size to match the kit, commonly 0.5 m or 1.0 m. Enable surface snap and offset by bounds so pieces rest on the hit surface.
3. Click in the viewport to place. Use the rotation value or stepped rotation to turn corners consistently.
4. Use Place In Front when surface placement is inconvenient. Use Replace Selected to swap a blockout piece while preserving its scene placement.
5. Enable paint mode only for repeated pieces. Set paint spacing to the module length to avoid overlapping walls.

Recommended replacement order

Order	Replace	Reason
1	Doorway and arch	Locks the main sight line and traversal clearance
2	Corner columns	Defines the kit's snapping rhythm
3	Wall spans	Fills measured gaps between anchors
4	Floor and stair trim	Finishes transitions after collision tests

Avoid scale drift

Prefer a correct grid and authored module dimensions over visually forcing pieces together with non-uniform scale. Non-uniform scale can create material stretching and collider mismatches.

07. Dress with reusable prefabs

Prefab Palette	PURPOSE	RESULT
Panels > Level Design	Browse categorized prefabs with favorites and controlled variation.	Reusable prop clusters placed quickly and consistently.

1. Open the palette and refresh if new prefabs were imported. Filter by category or name, then mark the lantern post and shrine cluster as favorites.
2. Enable Favorites Only while doing a focused pass. Select the lantern prefab and turn on placement.
3. Keep surface snap and bounds offset enabled. Use a grid size of 0.5 m for architecture-adjacent props.
4. Enable random variants if several assets share a variant group. Optional random yaw and scale are useful for organic props, but keep architectural props deterministic.
5. Place lanterns beside the entrance, benches against walls, and a shrine cluster on the central platform.
6. Use Replace Selected to test a different lantern or statue without rebuilding its placement.

Variation policy

Asset type	Yaw	Scale	Variants
Columns / arches	Fixed	1.0	Off
Lantern posts	Fixed or 180 degree steps	1.0	Optional
Rocks / debris	Random	0.8 to 1.2	On
Hero shrine	Fixed	1.0	Off

Prefab vs model

Use a model for one rendered mesh. Use a prefab when the placed item needs multiple objects, materials, colliders, lights, particles, scripts, or carefully authored child placement.

08. Create repeated patterns

Smart Duplicate & Array Panels > Level Design	PURPOSE Duplicate the selected source in linear, grid, radial, or spline layouts.	RESULT Editable repeated objects with a named generated group.
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1. Select one column beside the shrine and set Group Name to MC_ShrineArray. Keep replacement enabled.
2. Choose Linear, count 4, and spacing that matches the measured platform. Use local space when the source is rotated.
3. For floor tiles, choose Grid and set X/Y/Z counts and spacing. Keep the total below the tool's 2000-copy safety limit.
4. For candles around the shrine, choose Radial, use 8 copies, a 2 m radius, 360 degree arc, Y axis, and Face Outward.
5. For an existing spline, choose Spline, select the spline name, set start/end fractions, an optional offset, and align copies to the curve.

Choose the correct layout

Layout	Use it for	Key controls
Linear	columns, lamps, beams	count, vector spacing, rotation/scale step
Grid	tiles, shelving, seats	XYZ count and spacing
Radial	candles, stones, arena props	count, radius, arc, axis, outward facing
Spline	markers, signs, route props	spline, count, start/end, offset, alignment

Update safely

Generated names begin with the array group name. Recreate the pattern with the same group and replacement enabled to update it. Use Delete Generated Group when you want to remove the whole pattern.

09. Build the arrival road and fence

Spline Road & Fence Builder Panels > Level Design	PURPOSE Generate roads, fences, or evenly spaced prop lines along a scene spline.	RESULT Editable scene pieces following a curved route.
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1. Choose Add > Spline > Path Spline. Rename it MC_ArrivalPath. Add and move points so it starts outside the courtyard and ends at the doorway.
2. Edit points with W. Rotate spline points where bank or orientation controls are available. Keep point spacing broad enough for smooth curves.
3. Open Spline Road & Fence Builder, select MC_ArrivalPath, choose Road, and name the group MC_Road.
4. Set width 4.0 m, thickness 0.18 m, spacing near 1.0 to 2.0 m, and a small vertical offset. Select the road material, enable colliders, then Generate / Update.
5. Switch to Fence, set group MC_Fence, height 1.2 m, post size 0.14 m, rail thickness 0.12 m, and 2 rails. Regenerate after adjusting spline bends.
6. Switch to Prop Line, choose the lantern mesh, use 4 m spacing, align to spline, and generate a group named MC_RoadLanterns.

Curve quality

Symptom	Adjustment
Road looks segmented	Reduce segment spacing or soften spline control-point changes
Fence rails cut corners	Add a spline point near the turn and reduce spacing
Props face sideways	Toggle alignment or correct the mesh origin in Mesh Editor
Road floats / clips	Tune vertical offset and confirm spline point heights

Spline ownership

The builder reads the spline when you generate. Its output is a group of normal scene objects. After moving spline points, press Generate / Update again to rebuild the road, fence, or prop line.

10. Paint controlled natural detail

Scatter & Paint Panels > Level Design	PURPOSE Brush engine-owned static meshes onto surfaces with slope, spacing, and variation rules.	RESULT Natural detail that follows terrain and architecture.
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1. Select a rock or shrub mesh and activate the brush. Start with radius 2.0 m, density 0.5, stroke spacing 0.75 m, and minimum spacing 0.45 m.
2. Enable Align to Normal for rocks, Random Yaw, and Random Scale from 0.8 to 1.2. Use Keep Outside Surface to avoid embedding assets.
3. Set maximum slope near 45 to 50 degrees. Paint around the courtyard edges, not through the traversal lane.
4. Switch to Erase and remove objects from doorways, stair landings, camera sight lines, and combat space.
5. Use a smaller brush for intentional clusters near wall corners. Reserve high density for a few focal areas.

Three-pass dressing recipe

Pass	Brush	Goal
Large	rocks, large shrubs	Break the rectangular silhouette
Medium	grass clumps, debris	Connect architecture to the ground
Cleanup	erase mode	Protect movement and gameplay readability

Performance budget

Scatter creates many scene objects. Use minimum spacing, moderate density, and culling-friendly assets. Prefer the dedicated foliage actor for very large fields of repeated vegetation.

11. Organize the scene into layers

Level Layers & Visibility Panels > Level Design	PURPOSE Assign objects to named editor layers and manage visibility and locking.	RESULT A readable hierarchy and safe editing boundaries.
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Create these layers before the scene becomes crowded:

Layer	Contents	Typical state
Blockout	Original greybox and measurements	Hidden, unlocked when comparing
Architecture	Room shell, modular walls, stairs	Visible, locked during dressing
Roads	Spline, road, fences, route props	Visible
Gameplay	player start, triggers, nav volume	Visible during setup
Props	Prefabs and arrays	Visible
Foliage	Scatter-painted rocks and plants	Hidden during structural edits
Lighting	Directional, point, and spot lights	Visible

1. Create a layer, make it active, select related objects in the Hierarchy, and assign them to the active layer.
2. Lock Architecture after the shell is approved. Locked objects remain protected from accidental transform and deletion operations.
3. Hide Foliage while adjusting walls and splines. Show it again for final review.
4. Use the layer filter and object list to locate objects without scanning the entire hierarchy.

Layer naming

Use purpose-based names rather than asset types alone. A layer should answer why a set of objects is managed together: Blockout, Gameplay, Lighting, Roads, or ReviewOnly.

12. Save review viewpoints

Viewport Bookmarks & Navigation	PURPOSE	RESULT
Panels > Level Design	Capture and revisit editor camera positions with optional blending.	Fast, repeatable composition and gameplay reviews.

1. Frame the entrance and capture 01 Entrance Read.
2. Move to player eye height in the courtyard and capture 02 Player Route.
3. Capture 03 Shrine Hero from the intended reveal angle and 04 Top Down for layout review.
4. Set blend duration around 0.3 seconds. Visit each bookmark after structural or dressing changes.
5. Use Overwrite when refining a viewpoint, Rename for ordered labels, and Frame Selected when you need to locate the current object quickly.

Bookmark review questions

View	Ask
Entrance Read	Is the doorway obvious, and does the road point toward it?
Player Route	Are hazards and traversal gaps readable at gameplay height?
Shrine Hero	Does the shrine dominate without being blocked by props?
Top Down	Are density and empty space balanced across the level?

Bookmarks are review tools

Do not use only a free-flying beauty angle. Save at least one gameplay-height view and one top-down view so scale, collision, and composition errors are easier to catch.

13. Capture safe design alternatives

Level Snapshots & Variants	PURPOSE	RESULT
Panels > Level Design	Capture, compare, restore, duplicate, rename, and delete scene alternatives.	Reversible level-design decisions stored as assets.

1. With the greybox approved, capture Moonwatch_01_Blockout.
2. After modular replacement and spline work, capture Moonwatch_02_Architecture.
3. After props and foliage, capture Moonwatch_03_Dressed.
4. Duplicate the dressed variant as Moonwatch_04_OpenCombat. Restore it and erase extra foliage and props from the center.
5. Compare the dressed and open-combat variants from the same bookmarks. Restore the version with the clearest player route, then save the scene.

Variant operations

Operation	Use
Capture	Create a new snapshot from the current scene
Overwrite	Update an existing named variant
Restore	Replace the current working scene state with a variant
Compare	Review differences before choosing
Duplicate / Rename	Branch or clarify an alternative
Delete	Remove an obsolete variant asset

Save before restore

A restore changes the working scene. Save or capture the current state first when it contains work you may want to recover.

14. Validate and clean the level

Level Validation & Cleanup	PURPOSE	RESULT
Panels > Level Design	Scan the scene for unsafe values, missing assets, broken targets, and configuration problems.	A clean scene with actionable errors resolved.

1. Save the scene, then open Validation & Cleanup and run a scan.
2. Enable Info, Warning, and Error filters. Resolve errors first, warnings second, and informational cleanup last.
3. Use Fix only after reading the issue. Use Fix All Safe for items explicitly marked safe, such as certain naming or value corrections.
4. Rescan after fixes. Select each remaining issue to locate the affected object.

What the validator checks

Area	Examples
Identity	empty names and duplicate names
Transforms	invalid numbers and tiny scale
Assets	missing model, material, script, audio, particle, or behavior tree
References	missing target objects
Physics	invalid collider configuration
Scene hygiene	orphan empty objects and scene configuration problems

Validation is not playtesting

A clean scan proves that references and values are structurally valid. It does not prove that traversal, combat space, camera collision, lighting, or performance feels correct. Test those in Play mode.

SUPPORTING WORLD TOOLS

15. Integrate terrain, water, and foliage

These world systems sit outside the Level Design submenu, but they complete an outdoor level and should follow the same measured, layered workflow.

Terrain

1. Choose Add > Terrain. Keep the initial resolution modest while sculpting, and use the circular sculpt cursor to preview the affected area.
2. Shape broad slopes first, then refine. Avoid high-resolution edits until the silhouette and gameplay route are approved.
3. Assign terrain materials and verify the camera no longer clips through or incorrectly culls terrain near the surface.

Water and splines

4. Choose Add > Water > River for spline-driven water, or Lake/Ocean for broad surfaces. Edit the river spline and regenerate/update it after changing points.
5. Keep the river origin with the water plane. Review bends from above and at water level so width, flow direction, and UV continuity remain believable.

Foliage

6. Choose Add > Foliage Actor for large repeated vegetation fields. Use density, culling, random rotation, and the optional grass height randomizer to reduce repetition.
7. Use Scatter & Paint for authored clusters and the foliage system for large populations. Place both on the Foliage layer.

Performance workflow

Hide foliage while editing architecture, keep terrain sculpt resolution practical, use culling and instancing for vegetation, and review draw calls in the Profiler before increasing detail density.

END-TO-END RECIPE

16. Complete the Moonwatch Courtyard

Follow this compact sequence when repeating the workflow on a new level.

Done	Step	Tool	Deliverable
[]	1	Blockout	Floor, platform, stairs, doorway
[]	2	Measurement	Verified doorway, footprint, and traversal widths
[]	3	Alignment	Centered shrine and evenly spaced anchors
[]	4	Room Builder	Courtyard shell with entrance opening
[]	5	Modular Placement	Final wall, arch, column, and trim kit
[]	6	Prefab Palette	Lanterns, benches, and shrine cluster
[]	7	Smart Array	Columns, floor tiles, and radial shrine props
[]	8	Spline Builder	Arrival road, fence, and route lanterns
[]	9	Scatter & Paint	Edge rocks, shrubs, grass, and cleanup
[]	10	Layers	Blockout, Architecture, Roads, Gameplay, Props, Foliage, Lighting
[]	11	Bookmarks	Entrance, route, shrine, and top-down views
[]	12	Variants	Blockout, architecture, dressed, and open-combat states
[]	13	Validation	Zero unresolved errors and reviewed warnings

Final playtest

14. Enter Play mode. Traverse from the road to the shrine at walking and sprinting speeds.
15. Check camera collision near walls, stairs, fences, and foliage. Confirm decorative colliders do not block the player unexpectedly.
16. Review enclosed areas for correct darkness and shadow visibility distance. Check the scene from every bookmark.
17. Open the Profiler. Review frame time, draw calls, triangles, particle count, foliage count, water, and VRAM before adding more detail.
18. Return to Edit mode, make final corrections, rescan validation, save the scene, and capture the approved final variant.

FAST DIAGNOSIS

17. Troubleshoot common level-editing problems

Problem	Likely cause	Fix
Generated group duplicates	Replacement disabled or group name changed	Enable replacement and reuse the exact group name
Placement floats	Bounds offset, surface snap, or asset origin	Toggle bounds offset; correct the origin in Mesh Editor
Modules leave gaps	Grid does not match authored dimensions	Use the kit's true module size and reset non-uniform scale
Room door is misplaced	Wrong wall or offset	Select another door wall and regenerate
Array faces inward	Radial facing or local orientation	Toggle Face Outward or correct the source rotation
Spline road cuts a corner	Spacing too large or curve too sharp	Add a point and lower segment spacing
Scatter blocks gameplay	Painted through traversal space	Erase the route; lower radius or density
Cannot edit panel fields	Viewport camera currently owns the mouse	Release camera capture before using panel controls
Selection cannot move	Object or layer is locked	Unlock the object or its layer
Scene becomes slow	Too many unique objects or dense terrain/foilage	Use instancing, culling, lower density, and practical terrain resolution
Missing asset warning	Moved/deleted asset or stale reference	Reassign an engine-owned asset and rerun validation
Restore lost changes	Variant restored over unsaved work	Save or capture a variant before restoring another state

Reliable recovery order

Undo the last operation. If that is insufficient, restore the most recent saved scene. If the change was intentional but unsuccessful, restore the last approved Level Variant. Keep autosaves as an emergency fallback, not the primary workflow.

KEEP BESIDE THE EDITOR

18. Quick reference

Level Design panel map

Panel	Primary job	Produces / changes
Modular Placement	Kit-based placement	Models or prefabs in the scene
Prefab Palette	Reusable decorated placement	Prefab instances with variation
Room Builder	Captured room shell	Walls, floor, ceiling, posts, colliders
Scatter & Paint	Brush-based surface detail	Placed static mesh objects
Smart Duplicate & Array	Repeated patterns	Named linear/grid/radial/spline groups
Measurement & Ruler	Scale and clearance checks	Editor-only measurement guides
Validation & Cleanup	Scene integrity scan	Issues, safe fixes, and selections
Snapshots & Variants	Alternative level states	Restorable variant assets
Layers & Visibility	Organization and protection	Layer assignment, visibility, locking
Viewport Bookmarks	Repeatable camera review	Named editor camera viewpoints
Blockout & Shape Editing	Fast greybox geometry	Editable primitive groups
Object Alignment	Multi-selection layout	Aligned/distributed transforms
Spline Road & Fence	Curve-driven construction	Road, fence, or prop groups

Definition of done

The level is complete when the intended route is playable, scale is measured, repeated content is generated from named groups, objects are layered, critical views are bookmarked, alternatives are captured, validation has no unresolved errors, performance is within budget, and the final scene is saved.